BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 061 · Debugging and Nested Constructs

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does sequence mean in programming?

Q2. What is 4 + 3 * 2?
A. 10
B. 14
C. 20
D. 24

Q3. What is 2 ^ 3 in OCR ERL?

Q4. Join text '12' to '3'. Then convert '12' to an integer and add 3. Give both results. What should happen if the input is 'twelve'?

I Do · Read new knowledge

Key vocabulary

Syntax error — An incorrect language keyword or missing ENDIF breaks the grammar.

Logic error — A running program may produce an incorrect result because its steps are wrong.

Nested selection — Each IF within an IF needs the correct closing ENDIF.

Debugging cycle — Test, locate, correct and retest the fault.

Algorithm — A sequence of precise problem-solving instructions.

Trace table — A table tracking variable values during execution.

New learning

Syntax error: An incorrect language keyword or missing ENDIF breaks the grammar. For example, consider a membership checker.

Logic error: A running program may produce an incorrect result because its steps are wrong. For example, consider a sports bands.

Nested selection: Each IF within an IF needs the correct closing ENDIF. For example, consider a ticket discounts.

Debugging cycle: Test, locate, correct and retest the fault. For example, consider a club rules.

Worked example

A membership checker is being developed. Explain how syntax error and logic error are used. [4 marks]

1. Identify where this particular program needs control or values.

2. Syntax error [1]; An incorrect language keyword or missing ENDIF breaks the grammar. [1].

3. Logic error [1]; A running program may produce an incorrect result because its steps are wrong. [1].

4. Give one accurate point and a linked reason for each technique.

We Do · Practise with guidance

Guided 1

A sports bands is being programmed. Explain why syntax error matters. [2 marks]

Q3. Syntax error [1]; An incorrect language keyword or missing ENDIF breaks the grammar. [1]. Nested selection [1]; Each IF within an IF needs the correct closing ENDIF. [1]. Debugging cycle [1]; Test, locate, correct and retest the fault. [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Instructions execute in the specified order.

Q2. A — 10

Q3. 8; ^ denotes exponentiation in OCR ERL.

Q4. Joining the strings gives '123'. Converting '12' to an integer and adding 3 gives 15. Handle or reject 'twelve': it cannot be cast to an integer directly.

I Do / We Do pages · Guided practice

Q1. Syntax error [1]; An incorrect language keyword or missing ENDIF breaks the grammar. [1].

Q2. Logic error [1]; A running program may produce an incorrect result because its steps are wrong. [1].

Q3. Nested selection [1]; Each IF within an IF needs the correct closing ENDIF. [1]. Debugging cycle [1]; Test, locate, correct and retest the fault. [1].

You Do pages · Code First

```
octet = 260
if octet >= 0 AND octet <= 255 then
    print("Valid")
else
    print("Invalid")
endif
```

Expected result: Invalid

You Do pages · Independent questions

Q1. Syntax error [1]; Nested selection [1].

Q2. Logic error [1]; A running program may produce an incorrect result because its steps are wrong. [1]. Nested selection [1]; Each IF within an IF needs the correct closing ENDIF. [1].

Worked method: Syntax error [1]; An incorrect language keyword or missing ENDIF breaks the grammar. [1].

Guided 2

For a ticket discounts, explain how logic error is used. [2 marks]

Worked method: Logic error [1]; A running program may produce an incorrect result because its steps are wrong. [1].

Guided 3

A club rules needs nested selection and debugging cycle. Explain the roles of both. [4 marks]

Worked method: Nested selection [1]; Each IF within an IF needs the correct closing ENDIF. [1]. Debugging cycle [1]; Test, locate, correct and retest the fault. [1].

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Correct the logical operator so 260 is rejected.

```
octet = 260
if octet >= 0 OR octet <= 255 then
    print("Valid")
else
    print("Invalid")
endif
```

iterate

Question 1

State TWO concepts needed for this programming technique. [2 marks] Source: Original J277-style teaching question; not an OCR past-paper original.

Question 2

Explain how logic error and nested selection can help a ride height. [4 marks] Source: Original J277-style teaching question; not an OCR past-paper original.

Question 3

Explain the roles of syntax error, nested selection and debugging cycle when creating a reward scoring. [6 marks] Source: Original J277-style teaching question; not an OCR past-paper original.

RHS COMPUTING / OCR J277 / 2.1.2 / 2.2.1 LESSON 061 · DRAFT
BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

Exit Ticket · Answer before checking

An incorrect language keyword or missing ENDIF breaks the grammar.